

The 9×9 Wrapper Problem

A Self-Contained Statement: Definitions, Gates, Results, and What Remains Open

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Abstract

Composing the stratified vertical propagator with the disorder-resolved lateral propagator in a single resolvent requires both to speak the same 9-component language. They do not. This note defines every object involved, states every test applied to them and its exact outcome, and isolates what remains unknown. It is written to be recoverable without reference to any prior discussion: every symbol is defined where it first appears, every numerical claim names the script that produced it.

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1 Conventions and objects

1.1 Indices and ordering

Cartesian indices use the seismological convention of this project:

$$z = 0 \text{ (down)}, \quad x = 1 \text{ (right)}, \quad y = 2 \text{ (out)}, \quad (1)$$

right-handed. Voigt pairs are ordered

$$\alpha = 1, \dots, 6 \longleftrightarrow (p, q) = (0, 0), (1, 1), (2, 2), (1, 2), (0, 2), (0, 1), \quad (2)$$

i.e. (zz, xx, yy, xy, zy, zx). Shear components carry the *engineering* factor: the strain vector holds 2ε on the last three slots, while the stress vector holds plain σ . This asymmetry is the origin of the weight **W** below and is not optional — it is how the existing code is written.

Time convention is $e^{-i\omega t}$ throughout. Units are seismic: km/s, g/cm³, GPa, km.

1.2 The two state vectors

The *displacement–traction* 6-vector is

$$\mathbf{b} = (u_z, u_x, u_y, \tau_{zz}, \tau_{xz}, \tau_{yz})^T, \quad (3)$$

the natural state for a layered medium, since these six quantities are continuous across horizontal interfaces.

The *displacement–strain* 9-vector is

$$\boldsymbol{\psi} = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx})^T, \quad (4)$$

the natural state for a scatterer, since the single-site *T*-matrix acts on displacement and strain.

1.3 The two weights

$$\mathbf{W} = \text{diag}(1, 1, 1, 1, 1, 1, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}) \quad (9 \times 9), \quad (5)$$

$$\mathbf{J}_6 = \begin{pmatrix} \mathbf{0}_3 & \mathbf{I}_3 \\ -\mathbf{I}_3 & \mathbf{0}_3 \end{pmatrix} \quad (6 \times 6), \quad (6)$$

the canonical symplectic array of the thesis (its Eq. JDEF). Note $\mathbf{J}_6^T = -\mathbf{J}_6$ and $\mathbf{J}_6^2 = -\mathbf{I}_6$.

2 The three propagators

2.1 $\mathbf{P}^{\text{cf}}$ — the closed-form whole-space propagator

`cubic_scattering.horizontal_greens.exact_propagator_9x9(x, y, z,  $\omega$ , ref)` returns the 9×9

$$\mathbf{P}^{\text{cf}}(\mathbf{r}) = \begin{pmatrix} G & C \\ H & S \end{pmatrix}, \quad (7)$$

built from the Kupradze whole-space Green's tensor and its derivatives, where G is 3×3 (force $\rightarrow$ displacement), C is 3×6 (Voigt stress source $\rightarrow$ displacement), H is 6×3 (force $\rightarrow$ Voigt strain), and S is 6×6 (Voigt stress source $\rightarrow$ Voigt strain). Its *source* is a pair $(F_i, \Delta\sigma_{V\alpha}^*)$ — a body force and a stress-glut — and its *output* is ψ of (4). This object is treated here as ground truth; §4.1 states the evidence.

2.2 $\mathbf{G}_6$ — the layered displacement–traction Green's function

`GlobalMatrix.layered_greens.layered_greens_6x6(model,  $\omega$ ,  $k_x$ ,  $k_y$ ,  $j$ ,  $i$ )` returns a 6×6 mapping a source at interface j to the state $\mathbf{b}$ at interface i , at horizontal wavenumber (k_x, k_y) , for a plane-layered model. Two properties matter and are easy to miss:

1. Its **source is a state jump**, not a body force. Internally (`riccati_greens_psv`) a unit discontinuity is injected into the state vector, `rhs_p += I4`.
2. Its **basis is mixed**. The docstring of `riccati_greens_psv` gives it as $[u_x, u_z, \sigma_{zz}/(-i\omega), \sigma_{xz}/(-i\omega)]$: the traction half is scaled by $1/(-i\omega)$ relative to physical traction.

2.3 $\mathbf{A}_{\text{sys}}$ — the Fourier-domain system matrix

The layered problem marches as $\partial_z \mathbf{b} = \mathbf{A}_{\text{sys}} \mathbf{b}$ with (thesis Eq. AKDEF)

$$\mathbf{A}_{\text{sys}}(k_x, k_y) = \begin{pmatrix} 0 & -i\gamma k_x & -i\gamma k_y & a & 0 & 0 \\ -ik_x & 0 & 0 & 0 & b & 0 \\ -ik_y & 0 & 0 & 0 & 0 & b \\ -\rho\omega^2 & 0 & 0 & 0 & -ik_x & -ik_y \\ 0 & -\rho\omega^2 + \zeta k_x^2 + \mu k_y^2 & k_x k_y (\chi + \mu) & -ik_x \gamma & 0 & 0 \\ 0 & k_x k_y (\chi + \mu) & -\rho\omega^2 + \zeta k_y^2 + \mu k_x^2 & -ik_y \gamma & 0 & 0 \end{pmatrix} \quad (8)$$

where, and this is a genuine trap in the thesis,

$$\gamma = \frac{\lambda}{\lambda + 2\mu}, \quad a = \frac{1}{\lambda + 2\mu}, \quad b = \frac{1}{\mu}, \quad \zeta = \frac{4\mu(\lambda + \mu)}{\lambda + 2\mu}, \quad \chi = \frac{2\mu\lambda}{\lambda + 2\mu}, \quad (9)$$

with $\zeta - \chi = 2\mu$. These are *local* to (8); the thesis List of Symbols defines γ, ζ, χ as Chapter 2 contrast ratios, which do *not* apply. A footnote to this effect has been added to the recompiled thesis.

3 The wrapper under test

3.1 Its two operators

`layered_greens_9x9` constructs a 9×9 from $\mathbf{G}_6$ by sandwiching it:

$$\mathbf{M} = \mathbf{A}(\mathbf{k}) \mathbf{G}_6 \mathbf{B}. \quad (10)$$

The **receiver-side operator** $\mathbf{A}$ is `strain_from_displacement_traction`, a 9×6 mapping $\mathbf{b} \mapsto \boldsymbol{\psi}$, i.e. $(u, t) \rightarrow (u, \varepsilon)$:

$$\begin{aligned} \varepsilon_{zz} &= \frac{\sigma_{zz} - \lambda(ik_x u_x + ik_y u_y)}{\lambda + 2\mu}, & \varepsilon_{xx} &= ik_x u_x, & \varepsilon_{yy} &= ik_y u_y, \\ 2\varepsilon_{xy} &= ik_y u_x + ik_x u_y, & 2\varepsilon_{zy} &= \sigma_{yz}/\mu, & 2\varepsilon_{zx} &= \sigma_{xz}/\mu, \end{aligned} \quad (11)$$

with the displacements passed through unchanged. Note $\mathbf{A}$ depends on $\mathbf{k}$ and never sees k_z : it recovers ε_{zz} from σ_{zz} and the in-plane strains.

The **source-side operator** $\mathbf{B}$ is, in the shipped code, `traction_from_strain`: a 6×9 , wavenumber-*independent*, equal to $\begin{bmatrix} \mathbf{I}_3 & 0 \\ 0 & P_C \end{bmatrix}$ with C the isotropic Voigt stiffness and P the traction-extraction.

3.2 The corrections applied

Two modifications were derived and applied, giving what is called $\mathbf{M}^{\text{corr}}$ below:

$$\mathbf{M}^{\text{corr}} = \mathbf{A}(\mathbf{k}) [\mathbf{G}_6 \mathbf{Q}] \mathbf{B}^\dagger, \quad \mathbf{Q} = \text{diag}\left(1, -1, -1, \frac{i}{\omega}, -\frac{i}{\omega}, -\frac{i}{\omega}\right), \quad \mathbf{B}^\dagger = -\mathbf{J}_6 \mathbf{A}_{\text{src}}(-\mathbf{k})^T \mathbf{W}. \quad (12)$$

$\mathbf{Q}$ undoes the mixed basis on the source index; $\mathbf{B}^\dagger$ was derived by requiring the swap relation (14) below. $\mathbf{A}_{\text{src}}$ means $\mathbf{A}$ evaluated with the *source* interface's elastic properties (using the receiver's leaves an 83% residual).

4 The gates: exact statements and outcomes

Each gate is a necessary condition. Scripts named are in `scripts/` of `MultipleScatteringCalculations`.

4.1 GATE A — the invariant of the closed form

Statement. $\mathbf{W} \mathbf{P}^{\text{cf}}(\mathbf{r})$ is symmetric.

Origin. Measured, not assumed: the elementwise ratio P_{ab}/P_{ba} is an exact outer product w_a/w_b with $w = (1, 1, 1, 1, 1, 1, 2, 2, 2)$ — the Voigt engineering factor of (2).

Outcome. **PASS**, 4.2×10^{-18} at four separations; and 1.0×10^{-16} over twelve further random separations, so it is an identity rather than a coincidence.

4.2 GATE F — the same invariant, for the wrapper

Statement. In the homogeneous limit,

$$\mathbf{W} \mathbf{M}(k_x, k_y) \text{ is symmetric.} \quad (13)$$

Why this follows. $\mathbf{W} \mathbf{P}^{\text{cf}}(x, y, \Delta z)$ is symmetric at every (x, y) ; the 2-D transform over (x, y) is a linear combination with *scalar* coefficients; a linear combination of symmetric matrices is symmetric. Verified numerically as well: after transforming $\mathbf{P}^{\text{cf}}$ on a grid, $\mathbf{W} \tilde{\mathbf{P}}$ is symmetric to 6.4×10^{-16} across twenty wavenumbers.

Restriction. Licensed only in the homogeneous limit, where swapping source and receiver leaves the medium unchanged. For a genuinely stratified reference no within-matrix symmetry should be expected.

Outcome. $\mathbf{M}^{\text{corr}}$ **FAILS**, residual 0.40–0.81 across two frequencies and two slownesses.

4.3 GATE E — the swap relation

Statement.

$$\mathbf{M}(i \leftarrow j)(+\mathbf{k}) = \mathbf{W}^{-1} [\mathbf{M}(j \leftarrow i)(-\mathbf{k})]^T \mathbf{W}. \quad (14)$$

Outcome. $\mathbf{M}^{\text{corr}}$ **PASSES** at 9.7×10^{-16} .

What it cannot see. (14) relates two *different* matrices and is homogeneous of degree one in $\mathbf{M}$. It therefore constrains neither the overall scale (any constant multiplies both sides equally) nor the relation between the G, C, H, S blocks *within* one matrix. Both blind spots proved to be live.

4.4 GATE D — the reciprocity law of $\mathbf{G}_6$ itself

Statement. With $\mathbf{SD} = \text{diag}(1, -1, -1, -i\omega, +i\omega, +i\omega)$,

$$\mathbf{G}_6(i \leftarrow j)(+\mathbf{k}) = \frac{i}{\omega} \mathbf{SD} \mathbf{J}_6 [\mathbf{G}_6(j \leftarrow i)(-\mathbf{k})]^T \mathbf{J}_6 \mathbf{SD}. \quad (15)$$

Here $\text{diag}(1, 1, 1, -i\omega, -i\omega, -i\omega)$ is the $\sigma/(-i\omega)$ mixed basis applied uniformly to P–SV and SH, and $\text{diag}(1, -1, -1, 1, -1, -1)$ is the parity of the x, y components under $\mathbf{k} \rightarrow -\mathbf{k}$.

Outcome. **PASS**, 6.3×10^{-16} . Obtained by solving for the weight, not guessing: a joint rank-one fit over interface pairs, slownesses and frequencies has residual 4.8×10^{-15} with $\mathbf{J}_6$ and 0.97 with the identity, so the symplectic signature is required.

Trap. Recovering the SH entries requires $k_y \neq 0$. At $k_y = 0$ the SH channel decouples, components 2 and 5 vanish identically, and a fit returns an unconstrained value for them.

5 Components verified in isolation

Each of these was checked against an independent known answer. None is the defect.

Object	Check	Residual
A , Eq. (11)	Hooke's law on random plane waves	2×10^{-16}
G₆Q	GATE D, Eq. (15)	6.3×10^{-16}
P^{cf}	GATE A	4.2×10^{-18}
<code>_build_slab_kernels</code>	equals P^{cf} on the lattice	ratio 1.006, 1.001
inverse-FT pipeline	Weyl identity	ratio 0.98–1.03
G₆ solver	translation invariance, homogeneous	$\rightarrow 1.7 \times 10^{-5}$

For **A**: an arbitrary plane wave $u_i = U_i e^{i(k_z z + k_x x + k_y y)}$ gives ε and σ exactly from their definitions; forming **b** and applying (11) reproduces ψ to 2×10^{-16} over six random trials, including the non-trivial ε_{zz} row.

For the Weyl check: $\tilde{G} = (i/2k_z) e^{ik_z |\Delta z|}$ transformed by the same pipeline reproduces $e^{ikr}/(4\pi r)$, so the transform normalisation ($\text{ifft}2 \times N^2 \Delta k^2 / (2\pi)^2$) is correct and no stray 4π exists.

6 The negative results

These close off regions of the search space and are the substantive output.

N1. No B in the adjoint family satisfies E and F together. For $\mathbf{B} \in \{-\mathbf{J}_6 \mathbf{A}_{\text{src}}(-\mathbf{k})^T \mathbf{W}, \mathbf{S} \mathbf{J}_6 \mathbf{A}_{\text{src}}(+\mathbf{k})^T \mathbf{W}, \mathbf{J}_6 \mathbf{A}_{\text{src}}(+\mathbf{k})^T \mathbf{W}\}$ the shipped choice gives $E = 5.3 \times 10^{-16}$ with $F = 0.81$, and every variant improving F destroys E (best $F = 0.26$ at $E = 0.74$).

N2. GATE F reduces to a property of G₆ alone. If $\mathbf{B} = \mathbf{A}^T \mathbf{W}$ then

$$\mathbf{W} \mathbf{M} = (\mathbf{W} \mathbf{A}) \mathbf{G} (\mathbf{W} \mathbf{A})^T, \quad (16)$$

which is symmetric *if and only if* $\mathbf{G}$ is symmetric. So under that ansatz the question is not about **B** at all.

N3. No diagonal source correction makes G₆ symmetric. Requiring $\mathbf{G}_6 \text{diag}(q)$ symmetric gives 15 homogeneous equations in 6 unknowns. The null space does *not* exist: the smallest singular value ratio stalls at $4.9 \times 10^{-3} \rightarrow 1.8 \times 10^{-3}$ over $f = 6 \rightarrow 48$ Hz, the achieved symmetry is only $3\text{--}7 \times 10^{-2}$, and the fitted q drifts with frequency (0.48, 0.53, 0.56, 0.74). The system is inconsistent.

N4. A_{sys} is not the conversion needed here. Box 5.3 of the thesis gives the state-jump $\leftrightarrow$ body-force equivalence as $F = \mathbf{A}_{\text{sys}} S$ (derived and validated in `Mathematica/BoxFiveThree.wl`: quasi-Hamiltonian identity, eigenvalues, eigenbasis consistency, and the worked point-explosion

example all reproduced). Nevertheless every grouping $\mathbf{GQA}_{\text{sys}}^{-1}$, $\mathbf{GQA}_{\text{sys}}$, $\mathbf{A}_{\text{sys}}^{-1}\mathbf{GQ}$ leaves the symmetry residual at 1.4, no better than raw $\mathbf{G}$.

N5. The lateral and stratified propagators are not commensurable as built. In the homogeneous limit the ratio of the stratified spectral kernel to the verified lattice kernel varies by 10–14× across wavenumber and *grows* under damping, so it is not surface leakage. Discretising the stratified propagator onto the lattice improves this 3.8× (spread $9.8 \rightarrow 2.55$) but does not close it; the residual is a roughly constant factor near 12 with $\sim 200\%$ scatter.

7 What is open

Precisely one question, and it is small:

Given a 6×6 Green’s function whose source index is a *state jump* in the mixed basis of §2.2, what is the correct 6×9 source-side operator $\mathbf{B}$ such that $\mathbf{AG}_6\mathbf{B}$ reproduces the whole-space $\begin{bmatrix} \mathbf{G} & \mathbf{C} \\ \mathbf{H} & \mathbf{S} \end{bmatrix}$ of (7)?

By N2 the ansatz $\mathbf{B} = \mathbf{A}^T\mathbf{W}$ turns this into “is $\mathbf{G}_6\mathbf{Q}$ symmetric”, and by N3 no diagonal $\mathbf{Q}$ makes it so. Hence either $\mathbf{B} \neq \mathbf{A}^T\mathbf{W}$, or the source correction is not diagonal. Both remain open. The author of the underlying theory is better placed to answer this than further numerical search, which is why it is stated rather than pursued.

How to test any candidate. Run `scripts/gate_9x9_source_convention.py`. It executes gates A–F and exits non-zero unless A, E and F all pass. GATE F is proven valid (§4.2), so it will accept or reject a candidate immediately.

Reproducibility. Every number above comes from `scripts/gate_9x9_source_convention.py` (gates A–F), `scripts/composed_matvec.py` (N5, spectral), `scripts/discrete_pz_kernel.py` (N5, discrete), `scripts/gate_lateral_sweep_alg52.py` (lateral propagator), and `Mathematica/BoxFiveThree` (N4).